

0 – MongoDB: introduzione

1. Flat table

File

Home

Inserisci

Layout di pagina

Formule

Dati

Revisione

Visualizza

Sviluppo

🔍 Che cosa si desidera fare?

📁 Incolla

✂ Taglia

📄 Copia

📄 Copia formato

📌 Appunti

Calibri

22

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

🔍

2. DB normalizzato

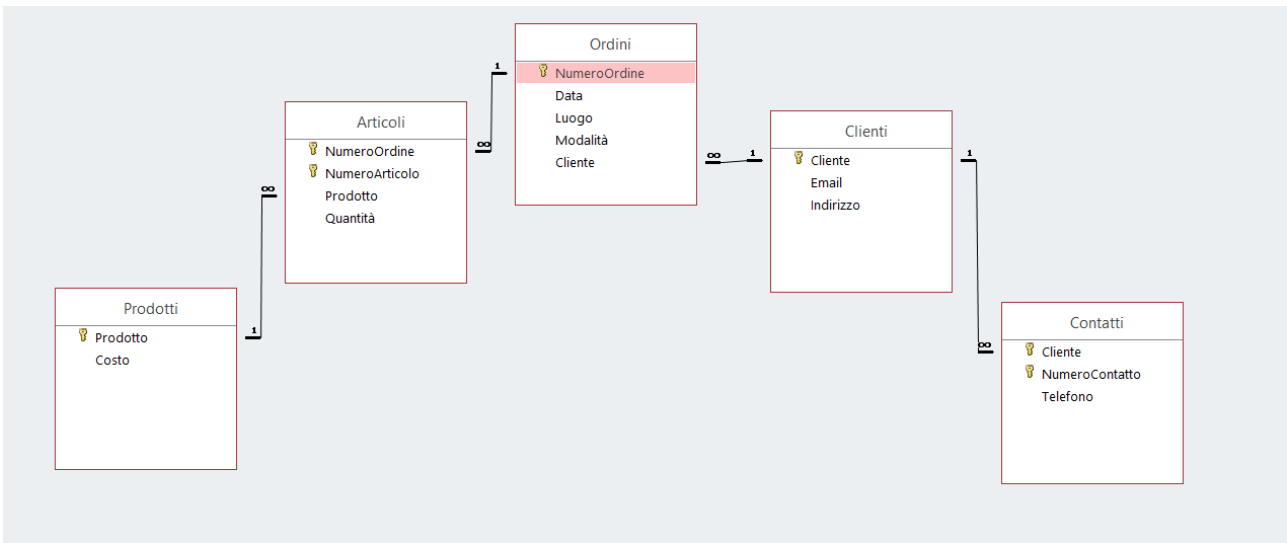
Ordini				
PK				FK
Nr.	Data	Luogo	Modalità	Cliente
1	23-gen-23	Milano	online	CLIA
2	4-dic-21	Lecco	negozio	CLIB
3	3-mar-22	Milano	negozio	CLIB
4	15-ago-21	Roma	negozio	CLIC
5	5-gen-23	Lecco	online	CLIC
6	25-dic-21	Roma	online	CLIA
7	5-set-21	Milano	online	CLIC
8	21-ott-21	Roma	online	CLIB
9	20-ott-21	Roma	online	CLIB
10	25-nov-22	Lecco	negozio	CLIC
11	9-gen-22	Milano	negozio	CLIA
12	12-giu-22	Roma	negozio	CLIC
13	6-set-21	Lecco	negozio	CLIB
14	4-lug-22	Roma	negozio	CLIB
15	6-dic-21	Milano	online	CLIB
16	11-lug-22	Roma	online	CLID
17	11-ott-22	Lecco	online	CLID
18	29-set-22	Lecco	negozio	CLIA
19	19-dic-21	Roma	negozio	CLIC
20	14-nov-21	Milano	negozio	CLID
21	8-feb-22	Firenze	online	CLIB
22	9-ago-21	Milano	online	CLIC
23	8-gen-22	Firenze	negozio	CLIA

Prodotti		
PK		
Prod.	Costo	
cartucce	20	
custodia	80	
laptop	900	
monitor	100	
printer	250	
tablet	400	

Contatti		
PK		
FK		
Cliente	Telefono	
CLIA	1	02-7050342
CLIA	2	333-123456
CLIA	3	328-6773111
CLIB	1	0341-452312
CLIC	1	06-2211546
CLIC	2	06-987236
CLID	1	055-370777

Clienti		
PK		
Cliente	Email	Indirizzo
CLIA	aaaaa@mail.it	via Bianchi 5, Milano
CLIB	bbbbbb@mail.it	via Rossi 8, Lecco
CLIC	cccccc@mail.it	via Verdi 9, Roma
CLID	dddddd@mail.it	via Neri 3, Firenze

3. Diagramma E/R



4. JSON

```
File Edit Selection View Go Run Terminal Help • vendite.json - Visual Studio Code
C: > Users > studente > Desktop > scuola 2022-2023 > TPS 5 > MongoDB > {} vendite.json > {} 0

58 {
59   "numeroOrdine": "4",
60   "data": "15/08/2021",
61   "negozio": "Roma",
62   "modalità": "negozio",
63   "articoli": [
64     {
65       "articolo": "cartucce",
66       "costo": "20",
67       "quantità": "3"
68     },
69     {
70       "articolo": "printer",
71       "costo": "250",
72       "quantità": "6"
73     }
74   ],
75   "email": "ccccc@mail.it",
76   "indirizzo": "via Verdi 9, Roma",
77   "telefoni": [
78     "06-2211546",
79     "06-987236"
80   ]
81 },
82 }
```

5. OLAP vs OLTP

Online analytical processing (OLAP) and online transactional processing (OLTP) are the two primary data processing systems used in data science.

- <https://corporatefinanceinstitute.com/resources/data-science/oltp/>
- <https://aws.amazon.com/it/compare/the-difference-between-a-data-warehouse-data-lake-and-data-mart/>

6. Database Normalization vs. Denormalization

- <https://medium.com/analytics-vidhya/database-normalization-vs-denormalization-a42d211dd891>
- <https://learn.microsoft.com/en-us/office/troubleshoot/access/database-normalization-description>

7. Horizontal Vs. Vertical Scaling

- <https://www.cloudzero.com/blog/horizontal-vs-vertical-scaling>